PERSONALIZED LEARNING PATH PREDICTION FROM STUDENT INTERACTION PATTERNS

A Mini Project Report Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

By

G Teja Swaroop – 2420030347 (Team Lead)

Sai Vivek – 2420030018

G Prasanth – 2420090072

Under the Esteemed Guidance of

[Name of the Applied Machine Learning for Text Analytics Faculty]



K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

The mini-project report entitled “Personalized Learning Path Prediction from Student Interaction Patterns” is a record of bona fide work carried out by G Teja Swaroop (2420030347), Sai Vivek (2420030018), and G Prasanth (2420090072) in partial fulfillment of the requirements for the award of the B.Tech degree at K L (Deemed to be) University. The work presented in this report is prepared for the Applied Machine Learning for Text Analytics project. All datasets, tools, code, research papers, and external sources used during implementation should be appropriately acknowledged. Experimental results will be included only after the model has been implemented and evaluated.

G Teja Swaroop – 2420030347

Sai Vivek – 2420030018

G Prasanth – 2420090072

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the mini-project report entitled “Personalized Learning Path Prediction from Student Interaction Patterns” is a bona fide work carried out by G Teja Swaroop (2420030347), Sai Vivek (2420030018), and G Prasanth (2420090072) under the supervision of [Faculty Guide Name] during the academic year [20__–20__], in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, [Faculty Guide Name], for the guidance, encouragement, and feedback provided during the planning of this project. We thank the faculty and staff of K L (Deemed to be) University for providing the academic environment and resources required to complete this mini-project.

We also acknowledge the maintainers of the EdNet dataset and the researchers and developers whose publications and software documentation are referenced in this report. Their work supports the study of student interaction patterns, educational data mining, machine learning, and explainable artificial intelligence.

G Teja Swaroop – 2420030347

Sai Vivek – 2420030018

G Prasanth – 2420090072

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	6
2.	Introduction	7
3.	Literature survey	8
4.	Problem Statement and Objectives	9
5.	Hardware and Software requirements	10
6.	Implementation	11
7.	Results & Discussion	12
8.	Conclusion & Future Scope	13
9.	References	14

1. ABSTRACT

Student interaction with online learning platforms provides useful evidence about learning behavior, progress, strengths, weaknesses, and engagement. Traditional learning systems often present a fixed sequence of materials to all learners and may not adapt to individual needs. This project proposes a machine-learning framework to predict suitable next learning activities or learning-path categories from historical student interaction patterns.

The proposed inputs include quiz attempts, assessment scores, time spent on learning materials, completed activities, content-access frequency, previous performance, and activity sequences. The EdNet dataset is identified as a candidate data source. The planned workflow includes data inspection, missing-value handling, numerical feature preparation where appropriate, behavioral feature engineering, and model training. Candidate algorithms include Random Forest, XGBoost, and Logistic Regression. SHAP-based explanations may be used to identify features that influence predictions.

The models are intended to be evaluated using accuracy, precision, recall, macro-F1 score, and a confusion matrix, depending on the final prediction target and class distribution. No measured performance results are available in the supplied abstract, so no model scores are claimed here. The intended outcome is an interpretable learning-path prediction framework that may help students follow suitable learning sequences and help educators identify topics or learners requiring additional support.

Team Name: [Enter team name]

Team Logo (optional): [Insert if available]

Team Members and Responsibilities:

1. G Teja Swaroop, 2420030347 – [Role/responsibility]

2. Sai Vivek, 2420030018 – [Role/responsibility]

3. G Prasanth, 2420090072 – [Role/responsibility]

Dataset/source: EdNet – https://github.com/riiid/ednet

2. INTRODUCTION

Online learning platforms record learner activities such as lesson access, quiz attempts, assessment outcomes, and time spent on content. These events can provide information about how learners interact with material over time. However, the volume and sequence of records make it difficult for educators to review every learner's progress manually and continuously.

A personalized learning path recommends a suitable next activity, topic, or module based on a learner's previous interactions and current performance. For example, repeated incorrect quiz attempts may indicate that revision material could be useful before a learner advances to a more difficult topic. Recommendations should support rather than replace educator judgement.

This project proposes a supervised machine-learning pipeline that transforms historical student interaction records into behavioral features and predicts a next learning activity or learning-path category. Candidate models are Logistic Regression, Random Forest, and XGBoost. SHAP-based explanations are proposed to make model predictions more understandable.

The intended users are students and educators. Students may use predictions to navigate learning content, while educators may use patterns to identify topics that appear difficult or learners who may need additional support. The final prediction target must be defined according to labels that can be reliably derived from the selected dataset.

The report reviews related work, defines the problem and objectives, lists hardware and software requirements, explains the planned implementation, and provides a structure for reporting measured results and future improvements.

3. LITERATURE SURVEY

1. EdNet dataset: EdNet is a large-scale dataset of student interactions with an online education platform. Its event-based records can support the analysis of question-solving behavior and learning activity sequences. It is a candidate source for deriving interaction features. Its records reflect a particular platform and population, so findings may not generalize automatically to other learning environments.

2. Knowledge Tracing: Corbett and Anderson introduced Bayesian Knowledge Tracing to model the acquisition of procedural knowledge over time. The approach provides a conceptual basis for estimating learner mastery from sequences of responses. It may require skill labels and sequential data that must be confirmed in the chosen dataset.

3. Deep Knowledge Tracing: Piech and colleagues used recurrent neural networks to model sequences of student interactions and predict performance. This work demonstrates the potential value of sequence-aware models. The present project initially considers classical models that are easier to benchmark and may be more practical for a mini-project.

4. Random Forest: Random Forest combines multiple decision trees and can capture nonlinear relationships among behavioral features. It is a candidate model for structured interaction data. Feature importance can provide a global view of model behavior, but does not fully explain every individual prediction.

5. XGBoost: XGBoost is a gradient-boosting framework for predictive modeling on structured data. It is included as a candidate model, subject to environment and data suitability. Appropriate validation and parameter tuning are necessary to avoid overestimating performance.

6. Logistic Regression: Logistic Regression provides a simple classification baseline and can be easier to interpret than complex models. Comparing it with tree-based models helps assess whether additional complexity offers a useful improvement.

7. SHAP: SHAP provides feature-contribution explanations for model predictions. It can support global summaries and individual explanations. Such explanations describe how a model uses features; they do not prove that a feature causes a learner’s outcome.

Research gap: This project plans to combine student-interaction-based prediction with an explainability layer, compare suitable models on a defined target, and examine whether the resulting predictions can be explained in a way that is useful to learners and educators.

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

How can historical student interaction patterns be used to predict an appropriate next learning activity or learning-path category and provide understandable explanations for that prediction?

Problem Description:

Traditional learning systems may present the same sequence of content despite differences in learner performance and engagement. Manually reviewing large volumes of interaction logs is time-consuming. A data-driven approach may help identify patterns associated with learning needs, provided that the prediction target, evaluation method, and limitations are clearly defined.

Objectives:

- Inspect the EdNet dataset or a documented alternative and summarize its fields, record counts, missing values, and target distribution.
- Define a reproducible target label for next-activity or learning-path-category prediction.
- Preprocess interaction data and engineer behavioral features from attempts, scores, activity completion, content access, and time-related fields where available.
- Train and compare Logistic Regression, Random Forest, and XGBoost where supported by the data and environment.
- Evaluate models using accuracy, precision, recall, macro-F1 score, and a confusion matrix.
- Use SHAP or an appropriate feature-importance method to examine model behavior.
- Discuss errors, class imbalance, privacy, fairness, and limitations.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Computer or laptop with a modern multi-core CPU.
- Recommended 8 GB RAM or more; larger data subsets may require additional memory.
- Sufficient storage for dataset files, processed data, notebooks, saved models, and figures.
- Internet access to obtain the dataset and packages; GPU is optional for classical machine-learning models.

Software Requirements:

- Python 3.x with Jupyter Notebook, Google Colab, or VS Code.
- pandas and NumPy for data handling and feature engineering.
- scikit-learn for preprocessing, model training, and evaluation.
- XGBoost package if the XGBoost model is implemented.
- SHAP for model explanations, if compatible with the selected model.
- Matplotlib and/or Seaborn for plots.
- EdNet dataset: https://github.com/riiid/ednet

6. IMPLEMENTATION

1. Data and prediction task

The EdNet repository is the dataset source identified in the abstract. The team should record the exact files or subset used, data usage conditions, number of records, available columns, and definition of one training example. The target must be derived without using information that would not be available at prediction time.

2. Preprocessing and feature engineering

Inspect data types, duplicates, missing values, and target distribution. Handle missing values using documented rules. Construct features such as prior quiz accuracy, attempt counts, recent performance, activity counts, content-access frequency, completion counts, and elapsed time where these fields exist. Encode categorical fields and scale numeric features where needed.

3. Model training and evaluation

Train Logistic Regression as a baseline and compare it with Random Forest and XGBoost where feasible. Use a documented validation strategy and record model parameters, random seed, software versions, and the metric used to select the final model. Fit learned preprocessing steps on training data only to avoid data leakage.

A. Project workflow and modules

• Data loading and exploration: [notebook/script name]

• Text/data cleaning and feature engineering: [notebook/script name]

• Feature extraction and model training: [notebook/script name]

• Evaluation, plots, and error analysis: [notebook/script name]

B. Machine-learning implementation

Describe the selected baseline and comparison models, the main parameters, and the purpose of each code module. Include relevant code excerpts or refer to the project repository.

C. Explainability / demonstration

If SHAP is implemented, describe the summary plot and selected individual explanations. If a dashboard is built, document the input validation, model loading, prediction display, and source files. Otherwise, document the notebook or command-line workflow.

7. RESULTS AND DISCUSSION

The supplied abstract describes a proposed system and does not contain completed experiments or measured model scores. The following items must be completed after implementation; no performance values should be invented.

Dataset summary

Dataset/subset: [fill in] Number of records/students: [fill in] Target classes: [fill in] Train/validation/test split: [fill in]

Model comparison

Model	Accuracy	Precision	Recall	Macro-F1
Logistic Regression	[TBD]	[TBD]	[TBD]	[TBD]
Random Forest	[TBD]	[TBD]	[TBD]	[TBD]
XGBoost (if used)	[TBD]	[TBD]	[TBD]	[TBD]

Figures and error analysis

Insert a target-class distribution chart, confusion matrix, and SHAP summary plot or equivalent feature-importance visualization. Explain each figure, identify common misclassifications, and discuss whether performance differs across classes. If a demonstration interface is built, include example predictions without exposing identifiable student information.

Limitations

Discuss class imbalance, incomplete or noisy interaction logs, differences between courses, changes in learner behavior over time, and possible bias in recorded data. Predictive performance alone does not prove that recommendations improve learning outcomes.

8. CONCLUSION AND FUTURE SCOPE

This project proposes a machine-learning framework for predicting personalized learning activities or learning-path categories from student interaction patterns. The planned workflow uses interaction data, behavioral feature engineering, and comparison of Logistic Regression, Random Forest, and XGBoost. SHAP-based explanations are proposed to make predictions easier to inspect. The planned evaluation uses accuracy, precision, recall, macro-F1 score, and a confusion matrix.

This report documents the proposed design based on the supplied abstract. It does not claim that the models have been trained or that any objective has been experimentally validated. After implementation, the team should add measured results and update the conclusion to reflect the evidence.

Future Scope

- Explore sequence-aware models or knowledge-tracing approaches if suitable sequential data are available.
- Test on more diverse courses or datasets to assess generalization.
- Investigate class-imbalance handling and probability calibration.
- Develop a student/educator dashboard after validating the model.
- Assess privacy, fairness, transparency, and recommendation impact through appropriate evaluation.

9. REFERENCES

[1] Riiid! Labs. EdNet dataset repository. https://github.com/riiid/ednet

[2] C. Piech et al. “Deep Knowledge Tracing.” Advances in Neural Information Processing Systems, 2015. https://proceedings.neurips.cc/paper/2015/hash/bac9162b47c56fc8a4d2a519803d51b3-Abstract.html

[3] A. T. Corbett and J. R. Anderson. “Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge.” User Modeling and User-Adapted Interaction, vol. 4, pp. 253–278, 1994. https://doi.org/10.1007/BF01099821

[4] L. Breiman. “Random Forests.” Machine Learning, vol. 45, pp. 5–32, 2001. https://doi.org/10.1023/A:1010933404324

[5] T. Chen and C. Guestrin. “XGBoost: A Scalable Tree Boosting System.” KDD, 2016. https://doi.org/10.1145/2939672.2939785

[6] S. M. Lundberg and S.-I. Lee. “A Unified Approach to Interpreting Model Predictions.” NeurIPS, 2017. https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[7] scikit-learn documentation. https://scikit-learn.org/stable/

[8] SHAP documentation. https://shap.readthedocs.io/

[9] pandas documentation. https://pandas.pydata.org/docs/

[10] NumPy documentation. https://numpy.org/doc/